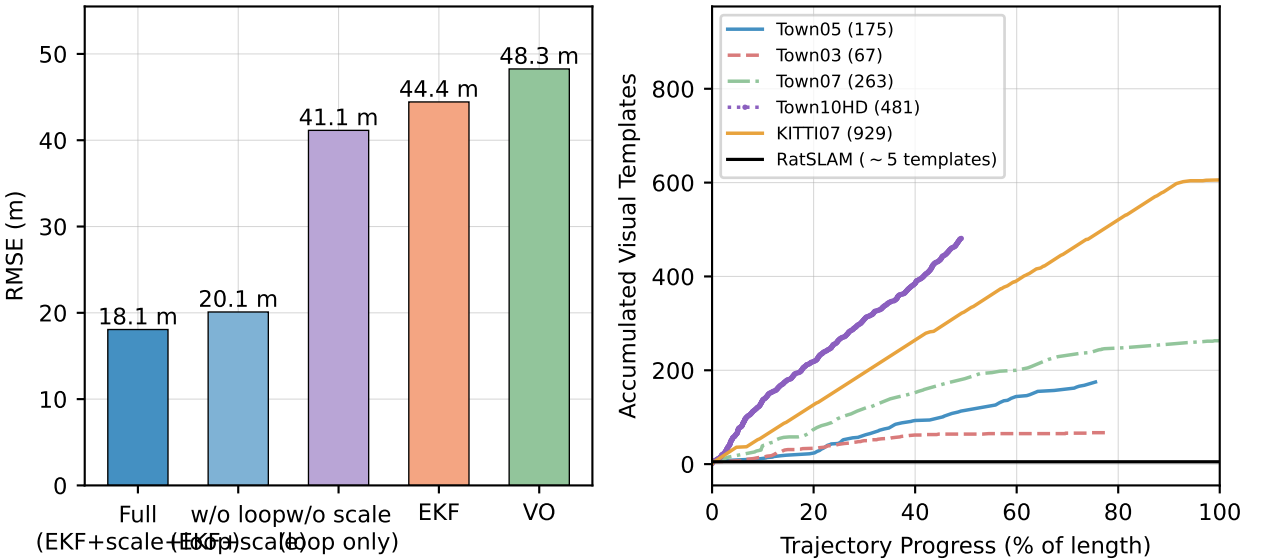
(a) Component Ablation on Town05 (486 m)
RMSE (m)
0
10
20
30
40
50
18.1 m
20.1 m
41.1 m
44.4 m
48.3 m
Full (EKF+scale+loop)
w/o loop (EKF+scale)
w/o scale (EKF+loop)
EKF only
VO
(b) Visual Template Growth (5 outdoor sequences)
Accumulated Visual Templates
0
200
400
600
800
Town05 (175)
Town03 (67)
Town07 (263)
Town10HD (481)
KITTI07 (929)
RatSLAM (∼5 templates)
0
20
40
60
80
100
Trajectory Progress (% of length)